

A Master long-case proforma and the 8–10 minute presentation

Use with every long-case card · Long case: 80 marks · 20 + 10 minutes

The department's shorthand

The order and wording on this card come from your department's own case sheets. Each subject card adds only what is special to its case. Learn the written shorthand too. Last year two long cases were full written case sheets.

Written	Meaning	Written	Meaning
c/o	complaints of	EOM	extraocular movements
h/o, No h/o	history of, no history of	Tn by GAT	tension (intraocular pressure) by Goldmann applanation tonometer
k/c/o	known case of	CCT	central corneal thickness
Ĉ	with	VH Gr IV (ND)	Van Herick grade 4, normal depth
⊕	present, positive	KPs	keratic precipitates
Ⓝ	normal	CPN (or NCP)	iris colour and pattern normal
s/o	suggestive of	RTL (3 mm)	pupil round, reacting to light, 3 mm
RE, LE, BE, B/L	right eye, left eye, both eyes, bilateral	RAPD	relative afferent pupillary defect
DM, SHTN, TB	diabetes mellitus, systemic hypertension, tuberculosis	PCIOL	posterior chamber intraocular lens
PR, BP, RR	pulse rate, blood pressure, respiratory rate	IMC (NS-3)	immature cataract, nuclear sclerosis grade 3
CVS, RS, P/A, CNS	cardiovascular system, respiratory system, per abdomen, central nervous system	PxF in ALC	pseudoexfoliation on the anterior lens capsule
S1 S2	first and second heart sounds	SS, TM, PAS	scleral spur, trabecular meshwork, peripheral anterior synechiae
NVBS	normal vesicular breath sounds	CDR, NRR, PPA	cup-disc ratio, neuroretinal rim, peripapillary atrophy
NFND	no focal neurological deficit	FR ⊕	foveal reflex present
E/D, BD	eye drops, twice daily	PH, DSph	pinhole, diopetre sphere
NIGNIP	no improvement with glasses or pinhole	6/12P	6/12, partial

Two traps. **VH** on the anterior chamber means Van Herick. On the department's fundus drawings it means vitreous haemorrhage, so write it out. **PVD** in the seniors' checklist means peripheral vascular disease.

Particulars and chief complaints

Particulars: name (Mr X), age, sex, occupation, place. The medical record number goes on the written sheet.

Patient	Write the opening line like this
New complaint	"Mr X, 35-year-old male, labourer by occupation, came with c/o pain, redness, watering and defective vision (LE) for the past 10 days."
Review patient	"Mr X, 70-year-old male, has come for regular follow-up."

- Put the eye in brackets after the complaints, and the duration as "for the past ...".
- With several complaints, give them in time order, each with its own eye and duration.
- Use the patient's complaint ("prominence of both eyes", "drooping of the left upper lid"), never your diagnosis.

History of presenting illness

Part	Say it like this
Start	"Patient was apparently normal ____ (days / months / years) back, then he noticed ____ in the ____ eye, which was:"
Onset	insidious / sudden / gradual in onset
Course	progressive / stationary in nature
Associations	associated / not associated with ____ (pain, redness, watering, photophobia, headache)

Part	Say it like this
Treatment so far	"He visited an ophthalmologist at his place, was found to have ____ and was started on _____. Then he came to our institution for further management."
Positives, then negatives	"H/o ____ (+)" lines first, then "No h/o ____" lines; the department stars each line

Elaborate each complaint:

Complaint	Ask
Defective vision	Eye, onset, progression; distance or near; pain; night blindness; change of glasses; coloured haloes
Pain	Onset, site, character, radiation; aggravating and relieving factors; eye movement; time of day
Redness, watering	Onset; pain, photophobia, discharge; foreign-body sensation; lid swelling
Double vision	Goes when either eye is closed?; horizontal or vertical; worse in which gaze; diurnal variation
Prominence of the eye	One or both; onset, progression; intermittent?; change with posture or straining; swishing sound
Drooping lid, squint	Congenital or acquired; progression; constant or intermittent; diurnal variation; glasses, patching

Negative history

- Each "No h/o ..." line rules out a named differential.
- Examiners ask "Why did you ask about ...?". Each card gives the reasons, and ask 1 of the five-asks method (below) shows how to say them.
- **The department's glaucoma screen:**
 - No h/o headache with vomiting and pain.
 - No h/o coloured haloes.
 - No h/o redness, watering, pain.
 - No h/o trauma to the eye.
 - No h/o frequent change of glasses.

Past, personal, family and treatment history

Part	Department wording or checklist
Past	"K/c/o DM / SHTN / TB / asthma / epilepsy for ____ years, on treatment ____" · similar complaints in the past · ocular surgery or trauma: eye, what, when, where
Seniors' drug and systemic checklist, every case	Sulpha allergy · other drug allergy · medical history · head injury · asthma · "CVD" (heart disease or stroke: ask both) · migraine · Raynaud's phenomenon · peripheral vascular disease · steroid use (type, duration)
Personal	"Takes mixed diet. Smoker / alcoholic for ____ years. Normal bowel and bladder habits."
Family	"No h/o similar illness in the family members"; ask about glaucoma, night blindness and consanguinity
Treatment	Drug, strength, frequency, eye, duration, compliance and side effects: "E/D timolol 0.5% BD (BE) for the past ____ years". Or: "Treated at a private clinic nearby with two eye drops, then referred to our institute."

General and systemic examination

Part	Department wording
General	"Patient is conscious, oriented, afebrile; moderately built and nourished; not anaemic, not icteric; no cyanosis, clubbing, generalised lymphadenopathy or pedal oedema."
Vitals	"PR ____/min, regular in rhythm, normal in volume · BP ____/____ mm Hg, measured in the left arm in sitting posture · RR ____/min."
Systemic	"CVS: S1 S2 heard, no murmurs · RS: NVBS, no added sounds · P/A: soft, no organomegaly · CNS: NFND."

Blocks added before the eye, when the case needs them:

- **Neck** (proptosis): a thyroid swelling that moves with deglutition.
- **Local examination of the eye** (corneal ulcer): lid closure, Bell's phenomenon and lagophthalmos. Then trichiasis, entropion, ectropion, blepharitis, meibomitis and a foreign body in the fornices.
- **Head posture and facial symmetry** (squint, palsy).
- **Cranial nerves I–XII** for palsies, as on the sixth nerve sheet.

Ocular examination — RE | LE table

Each row gives the right eye, then the left eye. The RE column shows a normal finding as the department writes it. The LE column shows an abnormal example, mostly from the department's or the model case sheets.

Row, in this order	Look for	RE — normal	LE — abnormal example
Visual acuity	Unaided → pinhole → best corrected (write the glass) → near	6/9, PH 6/6 6/9 C +0.50 DSph → 6/6 N6 at 33 cm	6/60 NIGNIP
Head posture	Face turn, head tilt, chin position; facial asymmetry	Normal	Characteristic face turn to the left side
Hirschberg test and ocular movements	Corneal reflexes, before dilating; ductions and versions	Reflexes symmetrical (orthophoric); EOM full	Abduction restriction ⊕
Lids and adnexa	Position, oedema, closure, lagophthalmos, trichiasis; lacrimal sac regurgitation	Ⓝ	Lower-lid oedema ⊕; inferior scleral show ⊕
Conjunctiva	Congestion (circumcorneal or conjunctival), chemosis, bleb; evert the upper lid	Clear	Circumcorneal congestion ⊕
Cornea	Clarity; oedema, opacity, ulcer, vessels, KPs; sensation; staining	Clear; sensation intact	Greyish-white 4 × 4 mm paracentral ulcer, well-defined margins, surrounding stromal infiltrate
Anterior chamber	Central and peripheral depth (Van Herick); cells, flare, hypopyon	VH Gr IV (ND)	ND; hypopyon ____ mm, mobile / not mobile
Iris	Pattern; new vessels; atrophy; nodules; synechiae; PxP; iridotomy patency	CPN	PxP ⊕ at the pupillary margin
Pupil	Size, shape, direct and consensual; swinging torch light test for RAPD, before dilating	RTL (3 mm)	4 mm, sluggish, grade 3 RAPD
Lens	Cataract and grade, PCIOL, aphakia, subluxation; PxP	Clear	NS-3; PxP in ALC
Intraocular pressure	Method, value, time; before gonioscopy and dilatation	Tn by GAT 14 mm Hg at 10 am	Tn by GAT 26 mm Hg at 10 am
Gonioscopy	Deepest structure seen per quadrant, on a cross; PAS, new vessels	SS in all quadrants (open)	TM all round; patent ostium and iridectomy superiorly
Fundus	Distant direct → direct → +90 dioptre lens → indirect (card B)	Media clear; disc and vessels appear normal; macula FR ⊕	CDR 0.8, NRR thinning ⊕ (superior, inferior), bayoneting ⊕, PPA ⊕
Fields (confrontation)	Each quadrant against your own field	Full in all quadrants	Defect ⊕ in the ____ quadrant

Tension: before gonioscopy (which lowers the reading) and before dilating (which raises it); always the method and the time. The department's tables list "Tn by GAT" straight after visual acuity. That is fine on paper if you measured it first.

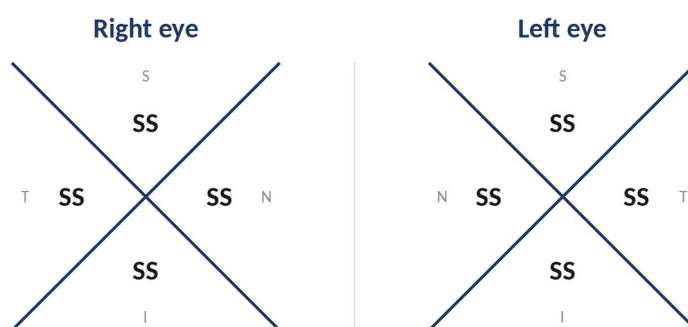
Van Herick: a thin slit 1 mm inside the temporal limbus, at 60°. Compare the peripheral chamber depth with the corneal thickness.

Grade 4 ≥ 1 (normal depth) $\cdot 3 \frac{1}{4} \cdot 2 \frac{1}{4} \cdot 1 < \frac{1}{4}$ (potentially occludable).

No slit lamp: the pen-torch method. Hold the torch on the temporal side. Whole iris lit = grade 4, normal depth. Two-thirds lit = grade 2, shallow.

Hirschberg test (before dilating): hold a pencil torch in the midline, 2 feet away. Ask him to look at it. Normally both corneal reflexes lie at the centre of the pupil. A reflex at the pupillary border means about 7° (14 prism dioptres) of deviation. At mid-iris it means about 15° (30 prism dioptres); at the limbus about 30° (60 prism dioptres). It is a rough estimate only.

Gonioscopy: draw one cross per eye and write the deepest structure seen in each quadrant. Name the system: "by modified Shaffer's grading", as the department writes. Last year's exam sheet:



Deepest structure seen in each quadrant — scleral spur all round: open angles (last year's exam sheet)

Fundus: describe it by card B. In a long case a normal fundus is written "Media clear; disc and vessels appear normal; macula FR ⊕".

Fields by confrontation:

1. Sit face to face with him, 60 cm apart.
2. His right eye fixes your left eye; the other two eyes are closed.
3. Bring your finger in from the periphery, midway between you, in each quadrant.
4. Compare his field with your own.

Summary

Part	Say it like this
Who and what	"Mr ___, a ___-year-old ___, came with ___ in the ___ eye for the past ___."
Course	"___ in onset, ___ in nature, associated with ___;"
History	"no history of ___; known case of ___."
Findings	"On examination, visual acuity ___ / ___; ___ [positives]; ___ [key negatives];"
Pressure	"intraocular pressure ___ / ___ mm Hg by ___ at ___."

Give only the positives that make the diagnosis and the negatives that exclude the differentials.

Complete diagnosis

Formula: eye → disease → type / stage / grade → aetiology → complications or status → other eye. Begin "My provisional diagnosis is ...". Say "probably of ... aetiology" when the cause is not proven.

Slot	Glaucoma sheet	Corneal ulcer sheet	Third nerve model sheet
Eye	Both eyes	Left eye	Right eye
Disease	primary open-angle glaucoma	corneal ulcer	third nerve palsy
Type / stage / grade	—	paracentral	pupil-sparing, infranuclear, incomplete
Aetiology	—	probably of bacterial aetiology	due to ischaemic microangiopathy
Complications or status	with pseudophakia, with intraocular pressure under control on medical management	in healing stage	—
Other eye	(both eyes)	state it if abnormal	state it if abnormal

More examples:

- "Bilateral axial proptosis, probably of thyroid-related orbitopathy, in active stage" (active or inactive by the signs).
- "Right eye: primary open-angle glaucoma, status post trabeculectomy with failed bleb and grade 2 nuclear sclerosis. Left eye: primary open-angle glaucoma and grade 2 nuclear sclerosis."

Differentials, investigations and management

- **Differentials:** 4–6, most likely first. For each, give the points for, the points against, and the sign or test that tells it apart (ask 3).
- **Investigations:** confirm → stage or grade → find the cause → baseline → fitness for surgery (ask 4). Central corneal thickness goes here, not in the clinical table.
- **Management:** aim → medical → laser → surgery → follow-up and counselling (ask 5). Give the trigger for each step up.

The five-asks method

Every case card answers the same five things, in the same places. Learn the method here once. Each card then fills it in for its own case.

No.	The ask	What you must be able to say
1	History, with a reason for every question	What each answer tells you. For each negative, what a "yes" would point to, and how to tell the look-alikes apart
2	The must-do examination, in order	How to do each step, how to record it, and the 2–4 viva questions it brings
3	Differentials	The clinching sign or test that tells each one apart
4	Investigations	The purpose of each: to confirm, to stage, to find the cause, baseline, fitness for surgery
5	Management	Aim → medical → laser → surgery → follow-up, with the trigger for each step up

Ask 1 — a reason for every history question

- **One question per row.** "No redness, watering or pain" is three questions with three different reasons.

- **Positive history:** onset, duration, progression, laterality, associated symptoms and treatment so far. Say what each answer tells you: the cause, how chronic it is, its stage, or a complication.
- **Negative history:** for each question, say what a “yes” would point to. Where your books give one, add the bedside sign or test that tells the look-alikes apart.
- **Past, drug, treatment, personal and family history:** give the consequence for diagnosis or for treatment.
- **Treatment history:** drug, strength, frequency, duration, compliance and side effects.

Worked example — the glaucoma negative history (the department’s starred lines, one per row):

Question	A “yes” would point to	How to tell them apart
Any coloured haloes?	Corneal epithelial oedema from a pressure rise: angle closure. Other causes of haloes: mucus on the cornea in conjunctivitis, incipient cataract, vitreous opacities, snow blindness, a tilted intraocular lens	Fincham (stenopaic slit) test: a slit is moved across the pupil. The glaucoma halo stays intact; the halo of incipient cataract breaks into its component colours
Any headache with vomiting and pain?	An acute angle-closure attack: pain, redness, headache, nausea and vomiting	Signs of the attack: a mid-dilated, vertically oval, poorly reacting pupil and a shallow chamber. Gonioscopy shows the closed angle
Any frequent change of glasses?	Raised pressure on the zonules and ciliary body impairs accommodation, so near vision suffers. Or he mistakes a field defect for poor vision	Perimetry: a field defect shows on the field chart
Any night blindness?	Defective dark adaptation , a feature of open-angle glaucoma. Miotic drops make it worse	Treatment history: is he using pilocarpine, a miotic?
Any injury to the eye?	Angle-recession glaucoma after trauma, usually in one eye	Gonioscopy shows the angle recession

Worked example — past and drug history:

Question	Why it matters
Asthma or chronic obstructive pulmonary disease?	Beta-blocker drops are contraindicated, so no timolol
Steroid use: which form, and for how long?	Steroid-induced glaucoma: open angle and raised pressure. Look for a posterior subcapsular cataract

Answering “Why did you ask...?” Give the reason. Then say what a “yes” would mean, and the test that would confirm it.

Q Why did you ask about coloured haloes?

Coloured haloes come from corneal epithelial oedema, which splits white light into its colours. A “yes” points to angle closure with a pressure rise. I would confirm it by Van Herick grading and gonioscopy, and use the Fincham test to exclude a cataract halo.

Ask 2 — the order of the examination earns marks

1. **Pupils before any drop.** Look for a relative afferent pupillary defect before the pupil is dilated. Look for iris new vessels at the pupillary margin before dilating, too. Count anterior chamber cells, measure a squint by Hirschberg and test colour vision before dilating.
2. **Tonometry before gonioscopy and dilatation.** Gonioscopy presses on the cornea and opens the angle, so a later reading is falsely low. Dilatation raises the pressure by 4–5 mm Hg, so a later reading is falsely high. Always note the method and the time.
3. **The disc after dilatation.** Examine the disc and nerve fibre layer through a dilated pupil. Use a +90 dioptre (D) or +78 D lens at the slit lamp. **Exception:** do not dilate an eye with a narrow or closed angle until a laser iridotomy is done.

Mnemonic — before you dilate: PART B. Pupillary reflex · Angle · Rubeosis · Tension · BCVA (best corrected visual acuity). The angle means both the anterior chamber angle and the angle of a squint.

The long-case order, step by step:

1. Visual acuity: unaided, pinhole, best corrected, near.
2. Head posture; Hirschberg test and ocular movements.
3. Pupils: direct and consensual reflexes; the swinging torch light test.
4. Torch, then slit lamp, undilated. Lids and adnexa → conjunctiva → cornea → anterior chamber (Van Herick) → iris → pupil → lens.
5. Intraocular pressure by applanation, with the time.
6. Gonioscopy, with the cross diagram.
7. Dilate, unless the angle is occludable. Look at the lens again, then the fundus: distant direct → direct → +90 D → indirect.
8. Fields by confrontation (the seniors’ addition).

Every step on a card has three parts:

- **Do:** how, in 1–3 lines, including what you tell the patient.
- **Record:** the department's shorthand.
- **Viva:** the 2–4 questions it brings, with their answers.

Worked example — the Van Herick step:

- **Do:** set a thin slit beam 1 mm inside the temporal limbus, at 60°. Compare the peripheral chamber depth with the corneal thickness.
- **Record:** "VH Gr IV (ND)", that is, Van Herick grade 4, normal depth.

Q Why do you grade the anterior chamber by Van Herick?

Van Herick grading is an initial guide to which patients need a detailed examination of the angle. Gonioscopy then shows the angle itself.

Q What does Van Herick grade 1 mean?

Grade 1 means the peripheral chamber depth is less than a quarter of the corneal thickness. The angle may be occludable, so I would do gonioscopy and not dilate yet.

Ask 3 — differentials, each with its clinching sign

- **Table:** Condition · Points for · Points against · How to tell apart. Give 4–6 rows, the most likely first.
- **The last column** names a sign you can show or a test result. Never write "clinical features".
- **Exclude-first line:** under the table, name the differential to exclude first, and how.
- **Say it like this:** "The differentials are A, which would show X, and B, which would show Y."

Worked example — primary open-angle glaucoma:

Condition	Points for	Points against	How to tell apart
Ocular hypertension	Pressure above 21 mm Hg on two readings; open angle	Glaucomatous disc and field	Disc, rim and field stay normal in ocular hypertension
Normal-tension glaucoma	Glaucomatous disc and field	Pressure above 21 mm Hg	Diurnal phasing: the pressure stays below 22 mm Hg; more disc haemorrhages and notching
Pseudoexfoliation glaucoma	Open angle; high pressure	No white material	White flakes on the pupillary margin and lens; Sampaolesi's line on gonioscopy
Chronic primary angle-closure glaucoma	Raised pressure; cupping	Deep chamber, Van Herick grade 4	Gonioscopy: closed angle. Indentation tells appositional closure from synechiae

Exclude first: angle closure, by gonioscopy, before you call any glaucoma open-angle.

Ask 4 — investigations, each with its purpose

Say them in this order:

1. **To confirm** the diagnosis.
2. **To stage or grade** it.
3. **To find the cause**, or a risk factor.
4. **Baseline** for follow-up.
5. **Fitness for surgery**.

For each test, know what you do, why, the expected finding and what it changes.

Worked example — primary open-angle glaucoma:

Purpose	Test	What it shows and what it changes
To confirm	Gonioscopy	An open angle. It excludes angle closure and secondary causes such as angle recession and pigment dispersion
To confirm and stage	Humphrey 24-2 or 30-2 field	Paracentral scotoma, arcuate scotoma, nasal step. The mean deviation (cut-offs -6 dB and -12 dB) stages the damage as mild, moderate or severe
To find a risk factor and correct the reading	Pachymetry: central corneal thickness	Goldmann applanation is accurate at 520 µm; the error is about 0.7 mm Hg per 10 µm. A thin cornea (below 500 µm) reads falsely low and is a risk factor. A thick cornea (above 570 µm) reads falsely high
Baseline for follow-up	Diurnal phasing; optical coherence tomography of the nerve fibre layer	A pressure swing of 8 mm Hg or more is significant. Superior and inferior nerve fibre thinning; the scan is the baseline for progression
Fitness for surgery	The tests the planned operation needs	Example: specular microscopy and biometry before cataract surgery in pseudoexfoliation (card G6)

Say it like this:

SAY IT

I will proceed with gonioscopy to confirm an open angle, and a Humphrey field to confirm and stage the damage. Pachymetry will correct the pressure reading. Diurnal phasing and a nerve fibre layer scan will give the baseline for follow-up.

Ask 5 — the management ladder, in the first person

Say it in this order: aim → medical → combinations → laser → surgery → follow-up and counselling. Give the trigger that moves you up each step. On every long-case card, two tables follow the ladder: “Drugs in this case” and “Laser and surgery in this case”. The full drug profiles are on the subject’s toolkit card (card GX for glaucoma).

Worked example — primary open-angle glaucoma:

SAY IT

My aim is to lower the intraocular pressure to a target pressure and stop further optic-nerve damage. The target pressure is the level at which further damage is unlikely.

First, medical treatment. I will start a prostaglandin analogue, latanoprost 0.005 per cent, one drop at bedtime. It lowers the pressure by increasing uveoscleral outflow. Its side effects are conjunctival hyperaemia, longer lashes and iris pigmentation. I avoid it in uveitis.

I will review him after two months. If the target is not reached, I will switch to another drug or add one. If two drugs are needed, I prefer a fixed combination. It gives simpler dosing, better adherence and less preservative toxicity. Its disadvantage is that the two drugs may need different times of day.

If he cannot use drops reliably, I will offer laser trabeculoplasty. Selective laser trabeculoplasty targets only the pigmented meshwork cells, so the tissue around them is spared heat. It is not suitable for uveitic, neovascular or angle-recession glaucoma.

The pressure may stay above target on three drugs, or the field may progress on maximal tolerated therapy. Then I will do a trabeculectomy. Its partial-thickness flap gives uniform pressure control and less hypotony. Its risks are hypotony with a flat chamber, bleb leak, bleb infection, endophthalmitis and cataract.

I will review him every three months, or monthly if the damage is advanced. Each time I will check the pressure and his compliance. I will repeat the field every six months, or every three months if advanced. I will repeat gonioscopy every year. I will counsel him that glaucoma is chronic, so the drops and the follow-up must continue. Lost vision does not return.

The triggers to step up, as the books give them for trabeculectomy:

- the target pressure is not reached on three topical drugs;
- the field progresses on maximal tolerated therapy;
- side effects or poor compliance;
- a baseline pressure above 40 mm Hg;
- one eye already lost, or follow-up likely to be difficult.

Seniors’ corrections — do these every time

- **EOM** row straight after visual acuity.
- **Van Herick grade** for every anterior chamber (“VH Gr IV (ND)”); the seniors circled it.
- **CCT out of the clinical table**: they struck the row out; it is an investigation.
- **“Fields (confrontation test)”** after the fundus, before the diagnosis.
- **The drug and systemic checklist** in every case; **complete the negative**: “No h/o headache **with vomiting and pain**”.
- **Gonioscopy**: name the grading (“by modified Shaffer’s grading”) and draw the cross.
- **Corneal ulcer**: site, size, shape, depth, colour, margin, floor, vascularisation, KPs, stromal infiltrate. Then sensation, staining, syringing, digital tension.
- **Cranial nerves**: name the muscles. The fifth nerve’s mandibular division supplies masseter, temporalis, medial and lateral pterygoids.

How to present it in 8–10 minutes

The older scheme gave the long case 20 + 10 minutes without saying how they are split. Prepare 8–10 minutes. If stopped, go straight to the summary and diagnosis (each card’s 2-minute say-it script).

Minute	Section	Key point
0–0.5	Particulars and chief complaints	One sentence: complaint, eye, duration
0.5–2.5	History of presenting illness	Onset, progression, associations; positives; negatives
2.5–3.5	Past, personal, family, treatment	Seniors’ checklist; drugs with strength and frequency
3.5–4	General and systemic examination	One sentence if normal; give the vitals

Minute	Section	Key point
4–7.5	Ocular examination, RE then LE	Brisk on normal rows; slow on the key structure
7.5–8	Special tests for this case	Gonioscopy, Hertel, ptosis measurements, cranial nerves
8–8.5	Summary	Positives and key negatives
8.5–9	Complete diagnosis	"My provisional diagnosis is ...", by the formula
9–10	Differentials, investigations, plan	One line each; then stop for questions

Saying the examination (minutes 4–7.5):

- Go row by row, in the table's order. In each row give the right eye, then the left eye.
- Group the normal rows in one sentence: "The lids, conjunctiva and cornea are normal in both eyes."
- Slow down on the key structure. Describe each finding, then say "suggestive of ...".
- Give the pressure with its method and time. Then give the special tests for this case.

Model opening — about 1 minute

SAY IT

My patient is a 35-year-old man, a labourer by occupation. He came with pain, redness, watering and defective vision in the left eye for the past 10 days.

He was apparently normal 10 days back. Then dust fell into his left eye while he was travelling on a two-wheeler. He had pain in that eye, sudden in onset and progressive in nature. It was associated with redness, watering, photophobia and defective vision. He was treated at a private clinic nearby with two eye drops, and then referred to our institute.

There is no history of contact lens wear, eye surgery, steroid drops or similar episodes in the past. He is not a known case of diabetes mellitus, hypertension, tuberculosis, asthma or epilepsy. He has no drug allergy. His personal and family history are not significant.

Model closing — the last 2 minutes

SAY IT

To summarise, my patient is a 35-year-old labourer with pain, redness, watering, photophobia and defective vision in the left eye. It began 10 days ago, after dust fell into the eye. He has no contact lens wear, steroid use or diabetes. Visual acuity is 6/6 in the right eye. In the left eye it is 6/60, not improving with glasses or pinhole. The left eye has lid oedema and circumcorneal congestion. It has a greyish-white, 4 by 4 millimetre paracentral corneal ulcer with well-defined margins. There is surrounding stromal infiltrate. The anterior chamber is of normal depth, with no hypopyon. The right eye and both fundi are normal.

My provisional diagnosis is left eye paracentral corneal ulcer, probably of bacterial aetiology, in healing stage. The main differential is a fungal ulcer, which would show feathery margins and satellite lesions.

I would take a scraping from the base and margins of the ulcer. I would send it for Gram stain, a 10 per cent potassium hydroxide mount and culture. I would treat according to the result, with atropine for the ciliary spasm. I would follow the ulcer until it heals.

How to answer any viva question

Examiners mark the shape of an answer as well as its content. Lead with the answer, never with background.

- **Keyword first.** Put the exact term at the start of the answer, in bold. The examiner then hears it in the first second. Say "**Bayoneting** is...", not "It is called bayoneting when...".
- **Use the keywords.** Each subject card has a line of the keywords the examiner listens for. Use each of those terms in your answers.
- **Definition first.** Start from the book's standard definition, with its key terms unchanged. Then give one line in plain words.
- **Plain English, professional register.** Short sentences, one idea each. Findings in the present tense: "The right eye shows...". Management in the first person: "I will start...". Numbers with their unit, method and time.

The shape for each kind of question:

The question asks for	Shape of the answer
A definition ("What is...?")	1. The standard definition, in the book's wording, key terms unchanged. 2. One line in plain words. 3. The 2–3 features that make the diagnosis
A classification	1. Name the system. 2. Give its grades in order. 3. Say which grade this patient has
Causes	1. Group them (congenital / acquired, ocular / systemic). 2. Most common first in each group. 3. The mnemonic, if your mnemonics file has one
Symptoms and signs	1. Symptoms first. 2. Signs in examination order: lids → conjunctiva → cornea → anterior chamber → iris → pupil → lens → intraocular pressure → angle → fundus

The question asks for	Shape of the answer
Difference between A and B	A table with the same rows for both: age, onset, symptoms, the key sign, the key test, treatment
Investigations	In order: confirm → stage or grade → find the cause → baseline for follow-up → fitness for surgery. Give the purpose of each
Management	Aim → medical → laser → surgery → follow-up and counselling, in the first person, with the trigger for each step up
A drug	Class → strength, dose and frequency → mechanism of action → adverse effects (ocular, then systemic) → contraindications
An operation or laser	Indications → principle → key steps (numbered) → advantages → disadvantages and complications
Complications	Grouped by time (during surgery / early / late) or by site (ocular / systemic). Most common and most serious first
"Why did you ask / check...?"	1. The reason, in one sentence. 2. What a positive answer would mean. 3. The test that would confirm it
Prognosis	Good factors, then poor factors

Two model answers

Q What is glaucoma?

Glaucoma is a chronic progressive optic neuropathy caused by a group of disorders leading to irreversible damage to optic nerves and corresponding visual field defects with or without high intraocular pressure. In plain words: the optic nerve is slowly and permanently damaged, and the field of vision shrinks to match. The diagnosis rests on the **optic disc**, the **visual field** and the **angle**.

Q Tell me about latanoprost.

Latanoprost is a **prostaglandin analogue**, used as 0.005% drops once at bedtime. It lowers intraocular pressure by **increasing uveoscleral outflow**. Side effects: conjunctival hyperaemia, longer lashes and iris pigmentation. Avoid it in uveitis.

The first answer has the definition shape: the book's definition, one plain line, then what makes the diagnosis. The second has the drug shape: class, strength and frequency, mechanism, adverse effects, contraindication.

Case-specific add-ons

What the typed formats ask for on top of this proforma:

Case	Extra examination	Cards
Glaucoma	Head and ocular posture; colour vision; skin naevi and phakomatosis (neurofibromatosis, Lisch nodules); patent ductus arteriosus with congenital rubella; lid hyperpigmentation, port-wine stain; conjunctival follicles, adrenochrome pigmentation, bleb; Krukenberg spindle; new vessels, PxF, iridotomy patency; retrolental cells; vertical and horizontal CDR, disc haemorrhage, vessel tortuosity, nerve fibre layer defects; gonioscopy diagram; automated perimetry	G1-G11
Cornea	Lid apposition, punctum, tear meniscus; evert the upper lid; corneal sensation, staining; each slit-lamp illumination; ulcer site, size, shape, edge, margin; labelled colour diagram	C1-C11
Proptosis	Axial or eccentric; pulsation; change with Valsalva; thrill, finger insinuation, retropulsion; auscultation for bruit; Hertel exophthalmometry; pressure in upgaze; forced duction	O1
Paralytic squint	Head tilt, face turn; primary and secondary deviation; diplopia charting, Hess chart; Park-Bielschowsky three-step test; double Maddox rod; past pointing	N1-N3, O3; charting on NX
Strabismus	Alternate cover; Krimsky; prism bar cover test in nine gazes; A or V pattern; accommodative convergence/accommodation ratio; Worth four-dot, Bagolini, Titmus; Bruckner	NX; N1-N3

Read more: Department case-sheet proforma (glaucoma, corneal ulcer, proptosis, uveitis, ptosis and sixth nerve sheets) · Typed case-presentation formats: glaucoma, cornea, proptosis, paralytic squint, strabismus · Aravind "Case sheet writing" p.649-667; 1.2; 4.4; 4.6; 4.8; 4.15; 4.17; 4.18 · Baidya p.12-18, 149-150, 154-155, 158-166, 196-198, 364, 557 · Namrata p.173-178, 181, 196, 205, 218, 352 · Mnemonics PDF: PART B

B Master fundus format

Use with every fundus card · Fundus: 60 marks (2 × 30) · 10 + 5 minutes per case

The instruction you may get

"Examine the fundus of this patient." Do exactly that, in the order below. Introduce yourself and take consent first.

The order

1. **Particulars and chief complaint** (shorthand as on card A), each eye with its own duration. Jot the history beside it (the history table below).
2. **Vision with pinhole (PH)**, right eye (RE) above left eye (LE): "6/24 → PH 6/18".
3. **Tonometry**: method, value, time (the department uses the non-contact tonometer).
4. **Anterior segment**: the four checks below, before dilating.
5. **Fundus description**, aloud, in the department's order below. It comes from the department's two written fundus write-ups: optic atrophy and diabetic retinopathy.
6. **Drawing**, colour-coded, every lesion labelled.
7. **Diagnosis** with grade or stage and the lens status, one line per eye.
8. **Plan**: eye + treatment, and the investigation that decides it.

Before you dilate — four checks

Point	Why	Record
New vessels on the iris (NVI)	Proliferative retinopathy, ischaemic vein occlusion: pupillary margin, high magnification	"NVI ⊕ at the pupillary margin"
Lens status	Part of the vision loss; a lens line in every diagnosis	"NS-2", "PCIOL"
Pseudoexfoliation	Weak zonules, glaucoma; on the department's vein occlusion sheet	"PxP in ALC"
Relative afferent pupillary defect (RAPD)	An ominous sign in vein occlusion; check before dilating	"Sluggish RTL", "RAPD ⊕"

Shorthand (as on card A)	Meaning
NS-2	nuclear sclerosis grade 2
PCIOL	posterior chamber intraocular lens
PxP in ALC	pseudoexfoliation on the anterior lens capsule
RTL	round, reacting to light

Q Why do you look at the iris before dilating?

Iris new vessels are looked for at the pupillary margin, under high magnification, before the pupil is dilated. In vein occlusion they are an important sign that decides the management.

Q Why check the pupils before dilating?

A **relative afferent pupillary defect** is an ominous sign in vein occlusion. Look for it before the pupil is dilated.

Q Why record the lens status?

The **lens status** explains part of the vision loss. Every diagnosis line carries it: "NS-2" or "PCIOL".

Q Why note pseudoexfoliation in a fundus case?

Pseudoexfoliation is linked with central retinal vein occlusion and glaucoma. It also means weak zonules if cataract surgery is needed later.

Fundus description — in this order

Step	Say it like this
Red glow	"On distant direct ophthalmoscopy at one arm's distance, there is a good red glow."
Media	"Direct ophthalmoscopy close to the face reveals clear media" (or "hazy media due to lens changes").
Disc: size, shape, colour, margins, cup-disc ratio, rim	"The disc is normal in size, vertically oval, pink, with well-defined margins, a cup-disc ratio of 0.3 and a healthy neuroretinal rim."
Vessels: origin, branching, arteriovenous (AV) ratio, calibre, crossings, sheathing	"The vessels arise from the centre of the disc, branching dichotomously, with an arteriovenous ratio of 2:3" (after the first branching); then attenuation, beading, AV crossings, sheathing.

Step	Say it like this
Background	Each lesion: number, colour, shape, size in disc diameters, margins, location → “suggestive of ...” (table below). Otherwise: “The background retina appears normal.”
Macula	Foveal reflex present or absent; oedema; hard exudates (circinate pattern?); scar.
Periphery	“Indirect ophthalmoscopy with a 20 dioptre lens shows the peripheries to be normal.”
Confirm	“On slit-lamp biomicroscopy with a plus 90 dioptre lens, the above findings were confirmed.”

The diabetic retinopathy write-up follows this order as far as the macula, where it stops. The optic atrophy write-up puts the +90 dioptre confirmation before the periphery.

Describe each lesion, then name it

Lesion	Say it like this
Hard exudates	“Yellowish, waxy lesions with distinct margins around the fovea, suggestive of hard exudates.”
Dot and blot haemorrhages	“Red, pinhead-shaped lesions in all four quadrants, suggestive of dot and blot haemorrhages.”
Flame-shaped haemorrhages	“Multiple linear red lesions temporal to the disc, suggestive of flame-shaped superficial haemorrhages.”
Cotton-wool spot	“A yellowish, fluffy lesion with indistinct margins, suggestive of a cotton-wool spot.”
New vessels on the disc	“A fine, lacy frond of vessels on the disc, suggestive of neovascularisation of the disc.”
New vessels elsewhere	“A whitish fibrous band with tufts of vessels, 1 disc diameter inferonasal to the disc, suggestive of neovascularisation elsewhere.”
Old laser marks	“Atrophic, hypopigmented chorioretinal scars along the arcade, suggestive of old laser marks.”
Pale disc (the department’s optic atrophy write-up)	“The disc is vertically oval, pale yellow and waxy in colour, with well-defined margins. Arteriolar attenuation is present.”

Say it — a normal fundus

SAY IT

Fundus examination of the right eye. On distant direct ophthalmoscopy at one arm’s distance, there is a good red glow. Direct ophthalmoscopy close to the face reveals clear media. The disc is normal in size, vertically oval and pink, with well-defined margins. The cup–disc ratio is 0.3, with a healthy neuroretinal rim. The vessels arise from the centre of the disc and branch dichotomously. They maintain an arteriovenous ratio of 2 is to 3, with no attenuation, crossing changes or sheathing. The background retina is normal, with no haemorrhages, exudates or pigment. The macula is normal, and the foveal reflex is present. On indirect ophthalmoscopy with a 20 dioptre lens, the peripheries are normal. On slit-lamp biomicroscopy with a plus 90 dioptre lens, the above findings were confirmed. The left fundus is the same.

Say it — an abnormal fundus (the department’s diabetic retinopathy write-up, with the last two steps added)

SAY IT

Fundus examination of the right eye. On distant direct ophthalmoscopy at one arm’s distance, there is a good red glow. Direct ophthalmoscopy close to the face reveals clear media. The disc is normal in size, vertically oval and pink, with well-defined margins. The cup–disc ratio is 0.3, with a healthy neuroretinal rim. The vessels arise from the centre of the disc and branch dichotomously. They maintain an arteriovenous ratio of 2 is to 3. Multiple yellowish lesions with distinct margins lie inferior to the disc and around the macula, suggestive of hard exudates. Multiple pinhead red lesions are seen in all four quadrants, suggestive of dot and blot haemorrhages. Multiple linear red lesions lie temporal to the disc, suggestive of flame-shaped superficial haemorrhages. Large, fluffy, yellow lesions lie in the superotemporal and superonasal quadrants, suggestive of cotton-wool spots. The macula shows oedema. On indirect ophthalmoscopy with a 20 dioptre lens, the peripheries are normal. On slit-lamp biomicroscopy with a plus 90 dioptre lens, the above findings were confirmed.

Then give the diagnosis line with its grade, the macular status and the lens status. The grading is on card R14.

Viva on the description

Q What does distant direct ophthalmoscopy show?

Distant direct ophthalmoscopy, at one arm’s distance, shows opacities in the media against the red glow. A greyish reflex suggests retinal detachment and a black reflex vitreous haemorrhage. A white reflex is leukocoria.

Q Where do you judge the arteriovenous ratio, and what is normal?

The **arteriovenous ratio** is judged after the first branching of the vessels. The normal ratio is 2:3.

Q Why are superficial haemorrhages flame-shaped and deeper ones dot-shaped?

Flame-shaped haemorrhages lie in the nerve fibre layer and follow its architecture; they come from the superficial precapillary arterioles. **Dot haemorrhages** lie in the inner retina, whose structures run perpendicular to the retinal surface.

Q What is a cotton-wool spot?

Cotton-wool spots are ischaemic infarcts of the nerve fibre layer. Axoplasmic flow is interrupted, so transported material builds up in the axons.

Q Why use both the direct and the indirect ophthalmoscope?

The **direct ophthalmoscope** gives a 15 times magnified, erect image of a 10–15° field, up to the equator. The **indirect ophthalmoscope** gives a stereoscopic view of a 35° field, out to the ora serrata. It sees better through hazy media.

Q Why confirm with the +90 dioptre lens?

Slit-lamp biomicroscopy with a +90 dioptre (D) lens is the best way to examine the disc and nerve fibre layer. It is done through a dilated pupil.

What to draw — Kanski's colour code

Two circles side by side, the right eye on the left. Write “RE” and “LE” above them (the department's circles are never labelled). Draw the disc on the nasal side, with the cup–disc ratio beside it. Every lesion gets a label and a pointer. Carry coloured pencils: red, blue, green, yellow, brown and black, and an eraser.

Colour	Standard code — draw this	What your department's sheets actually use
Red	Flat (attached) retina and haemorrhages	Haemorrhages of every kind; new vessels; one writer's fundus outline
Blue	Detached retina; retinal vessels (usually veins)	Disc outline; vessels, arteries and veins alike; labels and pointers
Red with a blue outline	Retinal breaks	Not used (no detachment last year)
Black	Pigment	Bony spicules, macular pigment, laser marks
Yellow	Exudates	Hard exudates; also disciform scar, sclerosed vessels, coloboma, macular flecks
Green	Vitreous opacities	Vitreous haemorrhage, fibrovascular proliferation, epiretinal membrane
Orange	Not in the code	Drusen

Your books add:

- arteries red, veins blue;
- the flap of a retinal tear blue;
- thinned retina in red hatches outlined in blue; lattice degeneration in blue hatches outlined in blue;
- drusen and chorioretinal coloboma yellow;
- sclerosed vessels black;
- silicone oil and gas green;
- choroidal lesions brown.

Where the department differs, use the standard colour and label it.

History to ask, if the examiner allows

Question	Why — what it rules in or out
Diabetes: since when, and how well controlled?	The duration of diabetes is the strongest predictor of retinopathy: about 25% at 5 years, 60% at 10 years and 90% at 15 years. Poor control raises the risk
Hypertension, raised lipids, kidney disease, anaemia?	They aggravate diabetic retinopathy. Hypertension, diabetes, smoking and raised lipids are also risk factors for vein occlusion
Laser or injections in this eye before? How many?	The treatment so far explains laser scars and guides the next step
Night blindness since childhood? Any relative affected?	Night blindness is the earliest and hallmark symptom of retinitis pigmentosa. Draw a pedigree chart, and note consanguinity as the department's sheets do
Chloroquine or phenothiazine (thioridazine) tablets?	Drug toxicity can mimic retinitis pigmentosa
Any similar symptoms in the other eye?	The other eye may be affected too

Complete diagnosis

- **Write “Diagnosis:”** with one line per eye and condition. Add the lens status and the grade, which the department's sheets leave out.
- **Example:** “Right eye high-risk proliferative diabetic retinopathy with clinically significant macular oedema · right eye pseudophakia”.

Differentials

Method: a table of Condition · How to tell apart, 2–4 rows, the most likely first. The last column names a sign you can show or a test result.

Worked example — haemorrhages and hard exudates in one fundus:

Condition	How to tell apart
Diabetic retinopathy	Microaneurysms, the earliest sign, with dot and blot haemorrhages and hard exudates; a history of diabetes
Central retinal vein occlusion	Flame and blot haemorrhages in all four quadrants, with dilated, tortuous veins: the “blood and thunder” appearance
Hypertensive retinopathy	Arteriolar narrowing and arteriovenous crossing changes come first; haemorrhages and cotton-wool spots come later
Drusen, mistaken for hard exudates	Round or oval, whitish or yellowish, with punched-out areas of choroidal or pigment epithelial atrophy. Hard exudates are waxy yellow, with distinct margins, among microaneurysms and haemorrhages

How I will proceed

Test	What it shows	Significance
Macular optical coherence tomography (OCT)	Retinal thickening, with cystic spaces in the inner nuclear and outer plexiform layers in diabetic macular oedema	Whether the oedema involves the central 1 mm subfield
Fundus fluorescein angiography (FFA)	Diabetic retinopathy: macular ischaemia; new vessels told apart from intraretinal microvascular abnormalities. Vein occlusion: the extent of capillary non-perfusion	Done after the acute haemorrhages settle in vein occlusion; shows conversion to the ischaemic type
B-scan ultrasonography	The retina behind hazy media, such as dense vitreous haemorrhage	Whether the retina is attached or detached; traction
Systemic work-up	Complete blood count, fasting and postprandial blood sugar, glycated haemoglobin, lipid profile, urea and creatinine, electrocardiogram	The basic tests in vitreous haemorrhage; diabetes control

How I will manage

- **Plan:** the eye, then the treatment. Examples: “(BE, both eyes) panretinal photocoagulation (PRP)”, “LE sectoral PRP”. Or “LE intravitreal bevacizumab, 3 doses”, written “Inj Avastin”.
- **Management:** say the ladder in the first person, as on card A (ask 5). The drug, laser and surgery profiles are on card RX.

Common fundus-description mistakes

- Naming a lesion without describing it (colour, shape, size, margins, number, location, then “suggestive of”).
- Skipping the red glow and the media; a cup-disc ratio without the rim; no AV ratio.
- “Macula normal” without the foveal reflex; no periphery; no 90 dioptre confirmation.
- Pupil and iris checked after dilating, so an RAPD or iris new vessels are missed.
- A diagnosis without its grade, stage or lens status.
- Unlabelled circles, eyes swapped; abbreviations aloud (“NVE” for new vessels elsewhere).

Quick recall

- **Order:** red glow → media → disc → vessels → background → macula → periphery → +90 dioptre confirmation.
- **Each lesion:** number, colour, shape, size in disc diameters, margins, location → “suggestive of ...”.
- **Before dilating:** iris new vessels, lens status, pseudoexfoliation, relative afferent pupillary defect.
- **Arteriovenous ratio:** 2:3, judged after the first branching.
- **Draw in red** the attached retina and haemorrhages; **in blue** the detached retina and veins.
- **Draw in black** pigment; **in yellow** exudates; **in green** vitreous opacities.

Read more: Department fundus case sheets (13 cases) · Department fundus write-ups (case-sheet proforma p.21, 30) · Aravind “Case sheet writing” p.649–650; 1.6; 1.7; 4.11; 6.1; 6.3; 6.5; 6.6; 6.8 · Baidya p.231–232, 249, 258, 273, 286, 318, 744–749 · Namrata p.173, 196, 218 · Kanski p.27

C Master short-case method

Use with every short-case card · Short cases: 60 marks (2 × 30) · 10 + 5 minutes each

The method, in one look

Every short-case card follows template B, in this order. This card teaches each heading, with a worked example: a pseudoexfoliation patient.

1. **The instruction you may get**, and do exactly that.
2. **Spot**: what you see first, said in one line.
3. **Keywords**: the 6–10 exact terms the examiner listens for.
4. **Focused examination**: numbered steps, each with Do, Record and its viva.
5. **Say-it description**, ending in the complete diagnosis line.
6. **History to ask, if the examiner allows**.
7. **Differentials**, each with the sign or test that tells it apart.
8. **How I will proceed**: investigations and their significance.
9. **How I will manage**: a short first-person ladder.
10. **Must know**: the definition first, then the classification and the key numbers.
11. **Viva questions and quick recall**.

The instruction you may get

Method: hear the instruction and do exactly that. Introduce yourself and take consent. Ask the age and complaint only if the examiner allows.

Examples:

- “Examine the anterior segment of this patient and measure the intraocular pressure.”
- “Examine the pupil and the lens of this patient.”

Spot

Method: inspect before you touch: head posture, face, both eyes compared, lids, deviation. Name the spot diagnosis in one line when asked.

Example: “White, flaky, dandruff-like material at the pupillary margin and on the anterior lens capsule, suggestive of pseudoexfoliation.”

Keywords the examiner listens for: pseudoexfoliation · four-point description · pupillary margin · three zones (target sign) · Sampaolesi’s line · poor mydriasis · phacodonesis · Goldmann applanation tonometry · gonioscopy · capsular tension ring

Focused examination — steps and their viva

Method: torch, then slit lamp, so the examiner sees each sign. Pupils and iris new vessels before any drop. Tension before gonioscopy and dilatation. Each step has **Do** (how), **Record** (the department’s shorthand) and 1–2 viva questions.

Step 1 — Inspect, then the pupils before any drop

Do: compare the two eyes with a torch. Check the direct and consensual reflexes, then the swinging torch light test for a relative afferent pupillary defect (RAPD). Look at the pupillary margin before any drop.

Record: “RTL (3 mm); no RAPD” (round, reacting to light).

Q Why do you look at the pupil before dilating?

The **pupillary margin** carries the hallmark flecks and shows loss of the pupillary ruff. **Poor mydriasis** is itself a sign of pseudoexfoliation.

Step 2 — Slit lamp, undilated

Do: grade the chamber by Van Herick. Examine the pupillary margin and iris, and retro-illuminate for transillumination defects. Look for iridodonesis and phacodonesis, and compare the chamber depth of the two eyes.

Record: “VH Gr IV (ND)” (Van Herick grade 4, normal depth); “PxP (pseudoexfoliation) at the pupillary margin, no iris atrophy”.

Q What are the four points to describe pseudoexfoliation?

The **four-point description** gives the colour of the material (white) and its nature (flaky, dandruff-like). Then its location (pupillary margin as tufts; anterior lens capsule as a disc-shaped deposit), and whether there is iris atrophy.

Q Why compare the chamber depth of the two eyes?

A difference in depth, or **phacodonesis**, points to zonular dialysis. Ultrasound biomicroscopy confirms its extent before cataract surgery.

Step 3 — Applanation tonometry

Do: consent, topical anaesthetic, fluorescein and a disinfected prism; both eyes (card G8).

Record: “Tn by GAT 24 mm Hg (RE), 16 mm Hg (LE) at 11 am”. That is tension by Goldmann applanation tonometry, right eye and left eye.

Q Why measure the pressure before gonioscopy and dilatation?

Goldmann applanation tonometry comes first. Gonioscopy presses on the cornea and opens the angle, so a later reading is falsely low. Dilatation raises the pressure by 4–5 mm Hg, so a later reading is falsely high.

Step 4 — Gonioscopy

Do: after tonometry and before dilating; topical anaesthetic and a disinfected lens; every quadrant (card G9).

Record: one cross per eye with the deepest structure seen in each quadrant. Note the pigment and Sampaolesi’s line beside it.

Q What is Sampaolesi’s line? Is it specific?

Sampaolesi’s line is a dark, dense, scalloped band of pigment on or in front of Schwalbe’s line. It is not specific: it also occurs in pigment dispersion syndrome and chronic inflammation.

Step 5 — After dilatation: the lens and the disc

Do: dilate, since the angle is open, and note how well the pupil dilates. Examine the anterior lens capsule, then the disc with a +90 dioptre lens.

Record: “Three zones on the anterior lens capsule (target sign); NS-2; CDR ____” (nuclear sclerosis grade 2; cup–disc ratio).

Q Describe the deposit on the lens.

The **three zones (target sign)** start with a central disc, absent in 20%. Next comes a clear zone, rubbed clean by the moving iris. Outside it lies a peripheral granular zone, seen only after dilatation.

Say-it description

Method: right eye, then left eye. Describe each finding, then say “suggestive of ...”. No abbreviations. End with the complete diagnosis line. The formula: eye → disease → type, stage or grade → aetiology → complications or status → other eye.

SAY IT

On slit-lamp examination of the right eye, the lids and conjunctiva are normal and the cornea is clear. The anterior chamber is of normal depth, Van Herick grade 4. There is white, flaky, dandruff-like material at the pupillary margin as tufts. There is a disc-shaped deposit on the anterior lens capsule, with no iris atrophy. This is suggestive of pseudoexfoliation. The pupil is round and reacting to light, and the lens shows grade 2 nuclear sclerosis. The left anterior segment is normal. Intraocular pressure by Goldmann applanation tonometry is 24 millimetres of mercury in the right eye. It is 16 in the left eye, at eleven in the morning.

My diagnosis is right eye pseudoexfoliation syndrome with raised intraocular pressure and grade 2 nuclear sclerosis.

History to ask, if the examiner allows

Method: 3–6 questions, one per row, each with what it rules in or out.

Question	Why — what it rules in or out
Any heart attack, stroke or high blood pressure?	Pseudoexfoliation is systemic: it is linked with myocardial infarction, stroke, hypertension and aortic aneurysm
Any glass blowing, or work with heat or infrared exposure?	A “yes” points to true exfoliation: the capsule itself splits into thin, clear scrolls
Any coloured haloes?	A “yes” points to angle closure: weak zonules let the lens move forward
Which drops, how often, since when?	One drug is usually not enough in pseudoexfoliation glaucoma; check compliance
Any eye surgery before?	After cataract surgery the intraocular lens may decentre late, because the zonules are weak
Asthma?	Beta-blocker drops are contraindicated, so no timolol

Differentials

Method: 2–4 rows. The last column names a sign you can show or a test result.

Condition	How to tell apart
Pigment dispersion syndrome	Young myopic men aged 30–50; Krukenberg spindle; mid-peripheral transillumination defects; a uniform dark pigment band on gonioscopy
True exfoliation	Clear scrolls split from the capsule itself; heat or infrared exposure; glaucoma is infrequent
Primary amyloidosis	Systemic and bilateral; fine whitish powdery deposits throughout the eye

How I will proceed

Method: 3–5 tests. For each, what it shows and what it changes.

Test	What it shows	Significance
Gonioscopy	Open or occludable angle; blotchy pigment; Sampaolesi's line	Open angle or angle closure (the angle is occludable in 9–18%)
Dilated disc examination and Humphrey field	Cupping with diffuse rim loss; a glaucomatous field defect	Pseudoexfoliation glaucoma rather than the syndrome alone
Diurnal variation of intraocular pressure	Wider swings than in primary open-angle glaucoma	Sets a realistic target pressure
Specular microscopy	Low endothelial cell density; altered cell size and shape	Fitness for cataract surgery
Ultrasound biomicroscopy, if phacodonesis	The extent of zonular dialysis	Plans the cataract surgery

How I will manage

Method: 3–6 first-person steps, each with its trigger. Each drug gets a one-line mechanism and its key adverse effect or contraindication. Each procedure gets one advantage and one disadvantage. Full profiles: card GX.

1. **Aim and approach.** I will treat aggressively, with frequent follow-up, because pseudoexfoliation glaucoma responds less well to drugs.
2. **Medical.** Latanoprost 0.005% at bedtime increases uveoscleral outflow; it darkens the iris and lengthens the lashes. Timolol 0.5% twice daily lowers aqueous production; I avoid it in asthma. One drug is usually not enough, so I will combine.
3. **Laser trabeculoplasty**, if the pressure is not controlled. Advantage: it is more successful than in primary open-angle glaucoma. Disadvantage: the effect does not last as long.
4. **Trabeculectomy**, if the target pressure is not reached or the glaucoma progresses; combined with cataract surgery when both are needed. Advantage: the partial-thickness flap gives uniform pressure control with less hypotony. Disadvantage: bleb leak, bleb infection and cataract.
5. **Cataract surgery with precautions.** Iris hooks or a pupil expansion ring for the small pupil. A dispersive viscoelastic for the endothelium. A capsular tension ring for weak zonules.
6. **Follow-up.** Yearly without glaucoma; every 3 months with glaucoma.

Must know

Method: the book's definition first, then related terms, the classification and the key numbers.

- **Pseudoexfoliation syndrome** is "an age-related systemic syndrome where mainly ocular tissues are affected through the gradual deposition of fibrillary white flaky material from the lens, mainly on the lens capsule, ciliary body, zonules, corneal endothelium, iris and pupillary margin resulting in secondary glaucoma." In plain words: white flakes from the lens settle all over the front of the eye and clog the drainage angle.
- **Glaucoma capsulare:** glaucoma in an eye with pseudoexfoliation; usually unilateral.
- **Who:** after 50, mostly in the seventh decade; men and women equally; commonest in Scandinavians.
- **40%** develop glaucoma or ocular hypertension. The angle is occludable in **9–18%**.
- **Hallmark:** flecks on the pupillary margin. **Most consistent sign:** the whitish powdery ring on the anterior capsule.
- **Pseudoexfoliation glaucoma versus primary open-angle glaucoma:** it starts later (over 60) and usually in one eye. The pressure is higher, often above 40 mm Hg, with wider swings. It responds less well to drugs.

Viva questions

Method: a real short-case card has 12–18, basics first. Here are four that come up in every pseudoexfoliation case.

Q What is true exfoliation?

True exfoliation is a lamellar delamination of the anterior lens capsule into thin, clear scrolls. It follows heat or infrared exposure, as in glassblowers. Glaucoma is infrequent.

Q Why are the zonules weak?

Zonular weakness comes from the material invading the zonular attachments; its proteolytic enzymes break them.

Q What precautions will you take at cataract surgery?

For the small pupil, iris hooks or an expansion ring. For the endothelium, a dispersive viscoelastic. For weak zonules, a chop technique and a **capsular tension ring**. After surgery, steroids given aggressively and for longer.

Q How does pseudoexfoliation glaucoma differ from primary open-angle glaucoma?

Pseudoexfoliation glaucoma starts later and is usually one-sided. The pressure is higher, with wider daily swings. It progresses faster and responds poorly to drugs.

Quick recall

- **Order:** pupils before drops → slit lamp undilated → tonometry → gonioscopy → dilate → lens and disc.
- **Four points:** colour · nature · location · with or without iris atrophy.
- **Three zones:** central disc (absent in 20%) · clear zone · peripheral granular zone (after dilatation).
- **Sampaolesi's line:** on or in front of Schwalbe's line; not specific.
- **40%** get glaucoma or ocular hypertension; occludable angle in 9–18%.

Glaucoma short cases were do-and-show

Last year most of the 12 glaucoma patients carried a task to perform in front of the examiner:

Task	Last year	Do it, then record
Applanation tonometry (card G8)	8 of 12	Consent, instruction, topical anaesthetic, fluorescein, disinfected prism; both eyes; eye, method, value, time
Gonioscopy (card G9)	8 of 12, every angle-closure glaucoma	After tonometry, before dilating; anaesthetic, disinfected lens; one cross per eye, deepest structure per quadrant; name the grading
Humphrey field (card G10)	3 of 12	Reliability first; defect on the pattern deviation plot; stage; match it to the disc

Treatment status was noted ("on anti-glaucoma medication" or not), so always state the current drops. Be ready for all three tasks on any glaucoma patient.

Timing — 10 + 5 minutes

- **Minutes 0–1:** instruction, consent, inspection.
- **Minutes 1–7:** examination.
- **Minutes 7–10:** description, diagnosis, differentials, plan.
- **Then 5 minutes of questions.**

The split is not official. If asked for the diagnosis early, give it at once.

Read more: Last year's glaucoma case list (tasks noted) · Department DigiNerve slides (pseudoexfoliation, Van Herick) · Cards G6, G8, G9, G10 · Baidya p.150, 155, 157, 167–169 · Namrata p.175–177, 203–207 · Aravind 4.11; 4.15; 4.18; "Case sheet writing" p.666

D Card index, checklist coverage map and toolkit list

★ = kept at JEH last year · 68 cards in 7 subject files, 6 of them toolkits · 21 cards marked ★

How to use this card

- **To find a checklist line**, look in the coverage map for its subject. The card and block named there answer it.
- **Each case card holds only what is special to its case.** The generic proforma, fundus format and short-case method are on cards A, B and C.
- **The full drug, laser and surgery profiles** are on each subject's toolkit card, the first card in each file.
- **Revise ★ cards first.** Joseph Eye Hospital (JEH) draws its exam cases from its own patients, so expect last year's pattern again.

Every card, by file

01_Glaucoma_Case_Cards

ID	Card	Type	★ last year
GX	Glaucoma treatment toolkit: every drug, laser and operation the glaucoma cards use	X	
G1	Primary open-angle glaucoma (POAG), including the glaucomatous optic disc	L+S	★ ×6
G2	POAG with trabeculectomy: bleb assessment, failed bleb	L	
G3	POAG with glaucoma drainage device (GDD)	L	
G4	Primary angle-closure disease: primary angle-closure suspect (PACS), primary angle closure (PAC) and primary angle-closure glaucoma (PACG), including after laser iridotomy and the acute attack. The deepest card: your thesis topic	L	★ ×4 (1 long case)
G5	Neovascular glaucoma (NVG), including rubeosis iridis	L+S	★ ×1
G6	Pseudoexfoliation syndrome and glaucoma	L+S	★ ×1 (one of G1's 6)
G7	Central retinal vein occlusion (CRVO) with POAG; hemi-CRVO with POAG	S	★ ×2
G8	Applanation tonometry: perform and record	T	★ task in 8
G9	Gonioscopy: perform, grade, draw	T	★ task in 8
G10	Humphrey field in glaucoma (last year's task; not on the checklist)	C	★ task in 3
G11	Other glaucomas the examiner may ask: childhood and secondary glaucomas (<i>extra</i>)	V	

02_Retina_Fundus_Case_Cards

ID	Card	Type	★ last year
RX	Retina treatment toolkit	X	
R0	Normal fundus: the perfect description	F	
R1	Proliferative diabetic retinopathy (PDR), including panretinal photocoagulation (PRP) marks and laser scars	L+S	★ ×3
R2	Non-proliferative diabetic retinopathy (NPDR) with diabetic macular oedema	L+S	
R3	CRVO: ischaemic, non-ischaemic, hemi-CRVO	L+S	★ ×2 (as G7)
R4	Branch retinal vein occlusion (BRVO); superotemporal BRVO (STBRVO)	L+S	★ ×2
R5	Age-related macular degeneration (ARMD), wet and dry	L+S	★ ×2
R6	Retinitis pigmentosa	L+S	★ ×4
R7	Retinal detachment	L	
R8	Hypertensive retinopathy	F	
R9	Stargardt disease	F	★ ×1
R10	Macular hole	F	
R11	Coloboma: iris, chorioretinal, disc	F	★ ×2
R12	Silicone-oil-filled eye	S	
R13	Fundus fluorescein angiography (FFA): NPDR, new vessels elsewhere (NVE), capillary non-perfusion (CNP)	C	

ID	Card	Type	★ last year
R14	Diabetic retinopathy chart: grading	C	
R15	Retinopathy of prematurity (ROP) chart	C	
R16	Fundus photographs: spotters	C	

03_Cornea_Case_Cards

ID	Card	Type	★ last year
CX	Cornea treatment toolkit	X	
C1	Fungal corneal ulcer	L+S	★ ×2 (1 long case)
C2	Corneal ulcer: the approach to infective keratitis (bacterial, including hypopyon ulcer; Acanthamoeba and viral keratitis as differentials)	L	
C3	Failed graft and the post-keratoplasty eye	L+S	★ ×1
C4	Keratoconus	L	
C5	Pseudophakic and aphakic bullous keratopathy	L	
C6	Opaque cornea, including adherent leucoma	L+S	
C7	Nummular keratitis	S	
C8	Corneal dystrophy	S	
C9	Corneal degeneration	S	
C10	Band-shaped keratopathy (BSK)	S	
C11	Atheromatous ulcer	S	

04_Lens_Uvea_Misc_Case_Cards

ID	Card	Type	★ last year
MX	Lens and uvea treatment toolkit	X	
M1	Ectopia lentis; Marfan syndrome with subluxated lens	L+S	★ ×1
M2	Dislocated lens in the vitreous	L+S	★ ×1 (fundus list)
M3	Anterior subluxation or dislocation of the lens	L+S	
M4	Dislocated or subluxated intraocular lens (IOL)	L+S	
M5	Developmental (zonular, lamellar) cataract and aphakia	S	★ ×1
M6	Traumatic cataract	S	
M7	Uveitis, including festooned pupil, iris nodules, synechiae	L+S	
M8	Hyphaema	S	
M9	Hypopyon and inverse hypopyon	S	

05_Nerves_Case_Cards

ID	Card	Type	★ last year
NX	Nerves treatment toolkit, including diplopia charting and the Hess chart	X	
N1	Third nerve palsy	L+S	★ ×1
N2	Fourth nerve palsy	L+S	
N3	Sixth nerve palsy	L+S	
N4	Optic atrophy	L+S	
N5	Anterior ischaemic optic neuropathy (AION): non-arteritic (NAION) and arteritic (AAION)	L+S	
N6	Facial nerve palsy, including lagophthalmos	L+S	

06_Oculoplasty_Case_Cards

ID	Card	Type	★ last year
OX	Oculoplasty treatment toolkit	X	
O1	Proptosis, including thyroid eye disease	L+S	
O2	Ptosis	L+S	
O3	Ocular myasthenia gravis	L+S	
O4	Anophthalmic socket	L+S	
O5	Contracted socket	L+S	

07_Viva_Charts_Instruments (built last)

ID	Card	Type	★ last year
V1	Visual fields: glaucoma and neurological	C	
V2	Imaging: optical coherence tomography, FFA and indocyanine green angiography, B-scan, Pentacam, specular microscopy, anterior-segment optical coherence tomography and ultrasound biomicroscopy, electroretinogram and visual evoked potential, computed tomography and magnetic resonance imaging	C	
V3	Instruments, by surgery	C	

Key:

- ★ = kept at JEH last year.
- ×n = how many of last year's 34 patients had it.
- For a task, ×n is how many of the 12 glaucoma patients carried it.
- Counts overlap: G6's patient is one of G1's six, and R3 counts the same two patients as G7.
- Types:**
 - L** long case; **S** short case; **L+S** one card with both versions.
 - F** fundus short case; **T** task you perform; **C** chart or picture.
 - X** treatment toolkit; **V** viva sheet of extra conditions.

Checklist coverage map

- Every line of the department's checklist, copied exactly as printed: abbreviations and spelling included.
- Card** answers the line. **Block** is the sub-heading in that card's "Short-case version", titled with the checklist wording.
- Whole card** means the card itself answers the line.
- A line that another subject's card answers shows that file in brackets.

Glaucoma

Checklist line	Long / short	Card	Block
POAG	Long	G1	Whole card
PACG	Long	G4	Whole card
POAG with trabeculectomy	Long	G2	Whole card
POAG with GDD	Long	G3	Whole card
Neovascular glaucoma (NVG)	Long	G5	Whole card
Pseudoexfoliative glaucoma	Long	G6	Whole card
Applanation tonometry	Short	G8	Whole card (task)
Gonioscopy	Short	G9	Whole card (task)
POAG disc / glaucomatous optic disc	Short	G1	Glaucomatous optic disc
CRVO with POAG	Short	G7	Whole card
Pseudoexfoliation	Short	G6	Pseudoexfoliation

Cornea

Checklist line	Long / short	Card	Block
Corneal ulcer	Long	C2	Whole card

Checklist line	Long / short	Card	Block
Fungal corneal ulcer	Long	C1	Whole card
Hypopyon ulcer	Long	C2	Its own section "Hypopyon ulcer"
Failed corneal graft	Long	C3	Whole card
Keratoconus	Long	C4	Whole card
Pseudophakic / aphakic bullous keratopathy	Long	C5	Whole card
Opaque cornea	Long	C6	Whole card
Nummular keratitis	Short	C7	Whole card
Corneal dystrophy	Short	C8	Whole card
Corneal degeneration	Short	C9	Whole card
Band-shaped keratopathy (BSK)	Short	C10	Whole card
Atheromatous ulcer	Short	C11	Whole card
Post-keratoplasty eye	Short	C3	Post-keratoplasty eye
Failed graft	Short	C3	Failed graft
Hypopyon	Short	M9 (misc file)	Hypopyon
Fungal corneal ulcer	Short	C1	Fungal corneal ulcer
Opaque cornea	Short	C6	Opaque cornea

Retina

Checklist line	Long / short	Card	Block
CRVO	Long	R3	Whole card
Ischaemic CRVO	Long	R3	Its own section "Ischaemic CRVO": how it is proved, its diagnosis line, its management
BRVO / STBRVO	Long	R4	Whole card
NPDR with macular edema	Long	R2	Whole card
PDR	Long	R1	Whole card
Wet ARMD	Long	R5	Whole card
Retinal detachment	Long	R7	Whole card
Retinitis pigmentosa	Long	R6	Whole card
Dry ARMD	Short	R5	Dry ARMD
Wet ARMD	Short	R5	Wet ARMD
NPDR	Short	R2	NPDR
PDR	Short	R1	PDR
Hypertensive retinopathy	Short	R8	Whole card (fundus)
BRVO / STBRVO	Short	R4	BRVO / STBRVO
CRVO	Short	R3	CRVO
Retinitis pigmentosa	Short	R6	Retinitis pigmentosa
Stargardt disease	Short	R9	Whole card (fundus)
Macular hole	Short	R10	Whole card (fundus)
Coloboma	Short	R11	Whole card (fundus)
Silicone-filled eye	Short	R12	Whole card
PRP marks	Short	R1	PRP marks
FFA: NPDR, NVE, CNP areas	Short	R13	Whole card (chart)
ROP chart	Short	R15	Whole card (chart)
Diabetic retinopathy chart	Short	R14	Whole card (chart)

Oculoplasty

Checklist line	Long / short	Card	Block
Proptosis	Long	O1	Whole card
Ptosis	Long	O2	Whole card
Ocular myasthenia gravis	Long	O3	Whole card
Anophthalmic socket	Long	O4	Whole card
Contracted socket	Long	O5	Whole card
Ptosis	Short	O2	Ptosis
Ocular myasthenia gravis	Short	O3	Ocular myasthenia gravis
Proptosis	Short	O1	Proptosis
Anophthalmic socket	Short	O4	Anophthalmic socket
Contracted socket	Short	O5	Contracted socket
Facial palsy	Short	N6 (nerves file)	Facial palsy

Nerves

Checklist line	Long / short	Card	Block
Third nerve palsy	Long	N1	Whole card
Fourth nerve palsy	Long	N2	Whole card
Sixth nerve palsy	Long	N3	Whole card
Optic atrophy	Long	N4	Whole card
AION	Long	N5	Whole card
Facial nerve palsy	Long	N6	Whole card
Third nerve palsy	Short	N1	Third nerve palsy
Fourth nerve palsy	Short	N2	Fourth nerve palsy
Sixth nerve palsy	Short	N3	Sixth nerve palsy
Facial nerve palsy	Short	N6	Facial nerve palsy
Optic atrophy	Short	N4	Optic atrophy
AION	Short	N5	AION

Miscellaneous

Checklist line	Long / short	Card	Block
Uveitis	Long	M7	Whole card
Ectopia lentis / Marfan syndrome with subluxated lens	Long	M1	Whole card
Anterior subluxation of lens	Long	M3	Whole card
Dislocated lens in vitreous	Long	M2	Whole card
Dislocated IOL	Long	M4	Whole card
Developmental / zonular cataract	Short	M5	Developmental / zonular cataract
Lamellar cataract	Short	M5	Lamellar cataract
Aphakia	Short	M5	Aphakia
Ectopia lentis	Short	M1	Ectopia lentis
Subluxated lens	Short	M1	Subluxated lens
Anterior subluxation of lens	Short	M3	Anterior subluxation of lens
Dislocated lens in vitreous	Short	M2	Dislocated lens in vitreous
Dislocated IOL	Short	M4	Dislocated IOL
Traumatic cataract	Short	M6	Whole card
Marfan syndrome with subluxated lens	Short	M1	Marfan syndrome with subluxated lens

Checklist line	Long / short	Card	Block
Pseudoexfoliation	Short	G6 (glaucoma file)	Pseudoexfoliation
Festooned pupil	Short	M7	Festooned pupil
Uveitis	Short	M7	Uveitis
Rubeosis iridis	Short	G5 (glaucoma file)	Rubeosis iridis
Hyphema	Short	M8	Whole card
Hypopyon	Short	M9	Hypopyon
Inverse hypopyon	Short	M9	Inverse hypopyon
Subluxated IOL	Short	M4	Subluxated IOL
Coloboma	Short	R11 (retina file)	The iris-coloboma part
Fundus photographs	Short	R16 (retina file)	Whole card (chart)
FFA	Short	R13 (retina file)	Whole card (chart)
ROP chart	Short	R15 (retina file)	Whole card (chart)
Diabetic retinopathy chart	Short	R14 (retina file)	Whole card (chart)
PRP marks	Short	R1 (retina file)	PRP marks

Toolkit list — where the full profiles are

- Each subject file opens with its toolkit card. It holds the full profile of every drug: class, strength and frequency, mechanism, adverse effects and contraindications.
- It also holds every laser and operation, with indications, key steps, advantages and disadvantages.
- A case card keeps a short drug table. It ends with “Full profiles: card GX” (or RX, CX, MX, NX, OX).
- A topic your three books do not cover is left off the toolkit.

Card and file	Drugs	Lasers and procedures	Surgery
GX • 01 Glaucoma (used by G1–G11)	Prostaglandin analogues, beta-blockers, alpha-2 agonists, carbonic anhydrase inhibitors (topical, oral), miotics, hyperosmotics; fixed combinations; drugs in the acute attack; newer drugs (<i>extra</i>)	Laser iridotomy, iridoplasty, laser trabeculoplasty (selective, argon), cyclophotocoagulation	Trabeculectomy with antifibrotics, combined surgery, drainage devices (valved and non-valved), non-penetrating surgery, cyclodestruction, lens extraction in angle closure; minimally invasive glaucoma surgery (<i>extra</i>)
RX • 02 Retina (used by R0–R16)	Anti-vascular endothelial growth factor (anti-VEGF) agents; intravitreal steroids and implants	Panretinal photocoagulation, focal and grid laser, barrage laser, photodynamic therapy, cryotherapy	Vitrectomy, scleral buckling, pneumatic retinopexy, tamponades (gases, silicone oil) and their removal
CX • 03 Cornea (used by C1–C11)	Antibacterials (fortified and commercial), antifungals (topical, oral, intrastromal, intracameral), antivirals, anti-Acanthamoeba agents, cycloplegics, steroids in graft rejection	Corneal scraping and culture, bandage contact lens, tissue adhesive, collagen cross-linking, ethylenediaminetetraacetic acid (EDTA) chelation, phototherapeutic keratectomy	Therapeutic and penetrating keratoplasty, deep anterior lamellar keratoplasty, Descemet's stripping endothelial keratoplasty (DSEK) and Descemet's membrane endothelial keratoplasty (DMEK), conjunctival flap, tarsorrhaphy, intracorneal ring segments
MX • 04 Lens, uvea, misc (used by M1–M9)	Topical, periocular and systemic steroids; cycloplegics; non-steroidal anti-inflammatory drugs; immunosuppressants and biologicals (<i>extra</i>)	Optical correction of aphakia and subluxation; neodymium-doped yttrium aluminium garnet (Nd:YAG) laser capsulotomy	Lensectomy (anterior, pars plana), capsular tension ring or segment, anterior-chamber, iris-fixated and scleral-fixated IOLs, IOL repositioning or exchange, paediatric cataract surgery, hyphaema wash-out
NX • 05 Nerves (used by N1–N6)	Steroids in optic neuritis and arteritic ischaemic optic neuropathy; lubricants in lagophthalmos	Occlusion, prisms, botulinum toxin; diplopia charting and the Hess chart (how to do, read and record them)	Squint surgery for palsies (recession-resection, transposition); lid procedures for lagophthalmos (tarsorrhaphy, gold weight, lateral tarsal strip)

Card and file	Drugs	Lasers and procedures	Surgery
OX · 06 Oculoplasty (used by O1–O5)	Thyroid eye disease: steroid regimens, selenium, radiotherapy. Myasthenia gravis: pyridostigmine, steroids, immunosuppressants, thymectomy	Ice-pack and neostigmine tests for myasthenia gravis	Ptosis surgery (Fasanella–Servat, Müller muscle–conjunctival resection, levator resection, frontalis sling and its materials); orbital decompression and the order of surgery in thyroid eye disease; orbital implants, dermis-fat and mucous-membrane grafts, prostheses

The viva file (07) has no toolkit. Its imaging and field cards point back to the subject toolkits for treatment.

Marks and timings

Head	Marks	Time for drills
Long case	80 (1 × 80)	20 + 10 min
Short cases	60 (2 × 30)	10 + 5 min each
Fundus	60 (2 × 30)	10 + 5 min (one case in the older scheme)
Refraction	30 (1 × 30)	10 + 5 min
Ward rounds	30 (5 × 6)	3 + 2 min per patient
Objective structured clinical examination	20 (5 × 4)	2 min per station
Dissertation	20	
Clinical total	300	
Viva, including competency	80 (60 + 20)	
Log book	20	
Aggregate	400; pass 200	

- **Marks:** the university's postgraduate practical scheme, as modified at the 64th Standing Academic Board on 25 November 2025.
- **Times:** from the older 300-mark scheme (last year's photo 1). Its marks no longer apply. Its timings are the only official ones found, so use them to time practice.

Last year's 34 cases at JEH

Category	Diagnoses (count)	Notes
Glaucoma (12)	POAG ×6 (one with pseudoexfoliation glaucoma; one both eyes) · PACG ×4 · neovascular glaucoma ×1 · CRVO with POAG ×1	Tasks written: applanation tonometry 8, gonioscopy 8, Humphrey field 3. Gonioscopy drawn as a cross: scleral spur in all four quadrants, both eyes.
Fundus (16)	Retinitis pigmentosa ×4 · proliferative diabetic retinopathy ×3 · age-related macular degeneration ×2 · superotemporal branch retinal vein occlusion ×2 · coloboma ×2 · hemi-CRVO with POAG ×1 · Stargardt disease (both eyes) ×1 · dislocated lens in the vitreous ×1	The hemi-CRVO patient was marked both fundus and glaucoma.
Anterior segment (5)	Fungal corneal ulcer ×2 · failed graft ×1 · subluxated lens (Marfan syndrome) ×1 · lamellar cataract / aphakia ×1	
Cranial nerve (1)	Third nerve palsy	On the main list only.
Total	34	

- **The iris coloboma:** the main list marks one coloboma patient as anterior segment ("iris coloboma"). The category sheets list her under fundus. By the main list, fundus is 15 and anterior segment 6.
- **Two case-sheet long cases:** one PACG and one fungal corneal ulcer were taken as full written case sheets. Practise writing a complete case sheet (card A) for both.

Read more: Department long- and short-case checklist · Last year's case lists (photos 2–7) · Older practical scheme (photo 1) · Tamil Nadu Dr M.G.R. Medical University postgraduate practical scheme, 64th Standing Academic Board, 25 November 2025 · Master prompt v4 §8